A COVERING OF A QUASI-COMPACT OBJECT CONTAINS A FINITE SUBFAMILY PRESENTING THE OBJECT BY EXTENSION BY ZERO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author invites the reader to skip the general case; the general argument is nevertheless present in the span and constitutes the counted proof, so the invitation was kept verbatim as part of the author's text.

Editorial note. The counted span cites six external items (lemmas, definitions, an equation), recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Quasi-compactness conditions for objects of a site and morphisms of coverings. Stacks Project, Sites, Lemma sites-lemma-conclude-quasi-compact (tag 09B1) and Definition sites-definition-morphism-coverings (tag 00VW). Supplies the refinement of a covering with finite index image used in the general case.
- Criterion for exactness by testing with Hom into sheaves of modules. Stacks Project, Homology, Lemma homology-lemma-check-exactness (tag 010B). Reduces the exactness of the displayed sequence to the corresponding statement after applying Hom.

Source reference. Original lemma 0934: sources/stacks-chow/tags.txt, source lines 11502–11502. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0D05: sources/stacks-chow/tags.txt, source lines 16300–16300. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00VT: sources/stacks-chow/tags.txt, source lines 1159–1159. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05AA: sources/stacks-chow/tags.txt, source lines 6843–6843. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 0G1V: sources/stacks-chow/tags.txt, source lines 20035–20035. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0G1L: sources/stacks-chow/tags.txt, source lines 20026–20026. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:sites-modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{U} = \{U_i \to U\}_{i \in I}$ be covering of $\mathcal{C}$. If U is quasi-compact, then there exist a finite subset $I' \subset I$ such that the sequence*

$$\bigoplus\nolimits_{i,i' \in I'} j_{U_i \times_U U_{i'}!}\mathcal{O}_{U_i \times_U U_{i'}} \to \bigoplus\nolimits_{i \in I'} j_{U_i!}\mathcal{O}_{U_i} \to j_!\mathcal{O}_U \to 0$$

is exact.

Proof. This lemma is immediate from Lemma 0934 if U satisfies condition (3) of Sites, Lemma 0D05. We urge the reader to skip the proof in the general case. By definition there exists a covering $\mathcal{V} = \{V_j \to U\}_{j \in J}$ and a morphism $\mathcal{V} \to \mathcal{U}$ of families of maps with fixed target given by $\mathrm{id} : U \to U$, $\alpha : J \to I$, and $f_j : V_j \to U_{\alpha(j)}$ (see Sites, Definition 00VT) such that the image $I' \subset I$ of α is finite. By Homology, Lemma 05AA it suffices to show that for any sheaf of $\mathcal{O}$-modules $\mathcal{F}$ the functor $\mathrm{Hom}_{\mathcal{O}}(-, \mathcal{F})$ turns the sequence of the lemma into an exact sequence. By (0G1V) we obtain the usual sequence

$$0 \to \mathcal{F}(U) \to \prod\nolimits_{i \in I'} \mathcal{F}(U_i) \to \prod\nolimits_{i,i' \in I'} \mathcal{F}(U_i \times_U U_{i'})$$

This is an exact sequence by Sites, Lemma 0G1L applied to the family of maps $\{U_i \to U\}_{i \in I'}$ which is refined by the covering $\mathcal{V}$. □